

Molecular, cellular and network mapping of brain structural deviations in patients with Post-COVID19 syndrome

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A B S T R A C T

Post-COVID-19 syndrome encompasses persistent cognitive, neurological, and psychiatric symptoms following SARS-CoV-2 infection, profoundly affecting global quality of life. Clarifying the neurobiological basis of these symptoms is vital for effective therapeutic interventions. This study utilized normative modelling of brain structure ("CentileBrain") to quantify subject-level deviations in cortical thickness, surface area, and subcortical volumes among 20 patients experiencing persistent fatigue following mild COVID-19, compared to 20 matched healthy controls. Group-level analyses on deviation scores revealed subtle yet distinct regional alterations in cortical thickness, specifically decreased thickness within orbitofrontal cortices and increased thickness in occipital/sensory cortices. Although at the individual regional level, the proportion of patients exhibiting infranormal or supranormal thickness values was relatively low (<35%) and comparable to controls, deviations frequently clustered within structurally connected circuits, affecting up to 50% more of patients. Spatial analysis of regional cortical thickness alterations correlated significantly with the constitutive expression patterns of TMPRSS2, an essential protein facilitating SARS-CoV-2 cellular entry. Canonical correlation analyses further identified specific cell-type distributions and neuroreceptor densities predictive of regional thickness changes, highlighting neurons and molecular targets associated with serotonergic, cannabinoid, cholinergic, and glutamatergic signalling pathways. Network-diffusion modelling constrained by a canonical structural connectome significantly outperformed null models based on permuted connectomes and Euclidean distance metrics, identifying posterior-parietal regions as probable initiation points ("seeds") for network-wide structural changes. Seed likelihood correlated positively with TMPRSS2 expression levels, suggesting that these posterior-parietal regions may be particularly susceptible to SARS-CoV-2 infection. This highlights a plausible mechanism where structural alterations could propagate through connected neural networks, although direct evidence of such propagation requires further investigation. These findings provide novel insights into potential mechanisms underlying neural circuit disruptions in post-COVID-19 fatigue and suggest avenues for therapeutic neuromodulation.

1. Introduction

Post-COVID-19 syndrome has rapidly emerged as a significant global health concern, characterized by an array of persistent cognitive, neurological, and psychiatric symptoms that occur following infection with SARS-CoV-2 (Davis et al., 2021; Nalbandian et al., 2021). Among these lingering symptoms, fatigue is notably prevalent and severely debilitating, adversely affecting daily functioning, productivity, and

overall quality of life for millions worldwide (Davis et al., 2021; Nalbandian et al., 2021; Cipolli et al., 2023). Despite extensive clinical research and public health awareness, the precise neurobiological mechanisms underlying these persistent symptoms, particularly fatigue, remain poorly understood. This gap in knowledge critically impedes the development of targeted therapeutic interventions aimed at alleviating the burden of post-COVID-19 syndrome (Nalbandian et al., 2021; Proal and VanElzakker, 2021; Peluso and Deeks, 2024).

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<https://doi.org/10.1016/j.bbih.2026.101292>

Received 4 March 2026; Received in revised form 21 June 2026; Accepted 21 June 2026

Available online 26 June 2026

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Magnetic resonance imaging (MRI) has played a central role in characterizing the neurobiological impact of COVID-19 across acute and post-acute stages. Beyond descriptive findings, MRI provides a non-invasive window into circuit-level brain organization, offering potential translational value for mechanistic understanding, patient stratification, and monitoring of post-infectious neurological sequelae (Li et al., 2025). Emerging evidence from neuroimaging studies has suggested that structural brain alterations might accompany even mild cases of COVID-19 infection (Li et al., 2025). These alterations include subtle but potentially significant changes in cortical thickness, surface area, and subcortical brain volumes within regions involving olfactory, frontoparietal or limbic circuits (Douaud et al., 2022; Lu et al., 2020; Alhazmi et al., 2023). However, existing findings are often inconsistent across studies, potentially due to methodological variability, small sample sizes, or the heterogeneity of patient populations. Moreover, traditional approaches typically rely on group-level analyses, which can obscure subtle individual differences in brain structure that may be clinically relevant. Consequently, a more nuanced approach capable of capturing individualized patterns of neuroanatomical deviation is necessary to better understand the underlying pathophysiological mechanisms (Ge et al., 2024; Lawn et al., 2024; Giacomel et al., 2025; Rutherford et al., 2023).

To address these limitations, our study employed a sophisticated normative modelling technique known as CentileBrain (Ge et al., 2024). This method leverages large, demographically diverse reference datasets - including tens of thousands of healthy individuals - spanning a wide age range to establish robust, population-informed models of brain structure. Using Gaussian process regression, CentileBrain generates centile-based predictions of cortical and subcortical anatomy for each individual, accounting for covariates such as age, sex, and intracranial volume. By comparing observed brain morphometry to these normative expectations, the approach yields individualized deviation scores (z-scores), allowing detection of subtle structural abnormalities that may be clinically relevant but diluted in group-level comparisons (Rutherford et al., 2022, 2023). In this study, we applied CentileBrain to quantify deviations in cortical thickness, surface area, and subcortical volumes among patients with persistent fatigue following mild COVID-19. This personalized framework would allow us the identification of heterogeneously distributed and circuit-convergent brain alterations, offering deeper insight into the neuroanatomical basis of post-COVID syndrome.

Given that structural brain alterations alone do not directly reveal the underlying biological mechanisms, we sought to contextualize our neuroanatomical findings within molecular, cellular, and network-level frameworks (Markello et al., 2022; Martins et al., 2021). To this end, we tested a series of interrelated hypotheses designed to illuminate potential drivers of the structural changes observed in PCS. First, we examined whether regional deviations in cortical thickness and surface area were spatially correlated with the expression of genes implicated in viral entry and neuroimmune signalling, such as TMPRSS2, FURIN, and NRP1 (Cantuti-Castelvetri et al., 2020; Moriyama et al., 2024), which are central to viral entry mechanisms, and TLR4, a key component of the innate immune system involved in pro-inflammatory signalling triggered by the viral spike protein (Asaba et al., 2024). This analysis was intended to explore whether direct viral neurotropism or associated entry mechanisms contribute to regional structural vulnerability.

Second, we investigated whether these anatomical changes were associated with the spatial distribution of distinct brain cell types and neurotransmitter receptor systems, drawing on normative transcriptomic and PET-based datasets (Markello et al., 2022; Martins et al., 2021). We focused in particular on systems previously implicated in fatigue and neuropsychiatric dysfunction, including serotonergic, cannabinoid, cholinergic, and glutamatergic pathways. This approach aimed to identify whether specific neural populations or signalling systems are preferentially involved in PCS, thereby suggesting possible neurochemical mechanisms or therapeutic targets for prioritisation.

Finally, we applied network-diffusion modelling using canonical structural connectivity data from the Human Connectome Project (Chopra et al., 2023; Bhattarai et al., 2022; Agosta et al., 2025; Parsons et al., 2021; Shafiei et al., 2023). This allowed us to test whether focal structural deviations might propagate along anatomical pathways, giving rise to the distributed and heterogeneous patterns of brain alterations seen in PCS. By identifying putative epicenters and characterizing their diffusion dynamics, this approach sought to offer insight into how localized pathology may result in widespread circuit dysfunction and the multifaceted clinical symptoms characteristic of post-COVID fatigue.

2. Methods

2.1. Study design and participants

As described elsewhere, this study employed a single-site observational case-control design. All participants, both PCS cases and recovered controls, were required to have had biologically confirmed SARS-CoV-2 infection at least three months prior to enrolment, consistent with the WHO clinical case definition of Post-COVID-19 condition (Soriano et al., 2022). Individuals in the PCS group were characterised by persistent symptoms lasting ≥ 3 months after infection, whereas recovered controls were required to be fully asymptomatic, with complete resolution of all COVID symptoms. We included individuals with mild acute COVID-19 only, to minimize potential confounding effects of hospitalisation, hypoxia, or severe systemic inflammation on brain perfusion and metabolism, and to ensure that any observed group differences were not attributable to the neurological sequelae of moderate-severe disease. All participants were required to have had biologically confirmed SARS-CoV-2 infection prior to enrolment. Confirmation was based on either a positive reverse-transcription polymerase chain reaction (RT-PCR) test from a nasopharyngeal swab or documented serological evidence of prior infection, in accordance with UK national testing guidelines at the time of acute illness. Participants with only clinically suspected but biologically unconfirmed COVID-19 were not included. Recovered healthy control participants were required to have a history of biologically confirmed SARS-CoV-2 infection, followed by complete resolution of all acute and post-acute symptoms. To minimize the risk of including individuals with subtle or subclinical post-COVID manifestations, control participants were screened using the same standardized clinical instruments applied to the PCS group, including the Fatigue Assessment Inventory. Only individuals scoring below established clinical thresholds, and reporting no persistent fatigue, cognitive complaints, affective symptoms, autonomic dysfunction, or post-exertional malaise at the time of assessment, were included as healthy controls. In addition, control participants underwent the same cognitive assessment battery and clinical interview as PCS participants, and individuals demonstrating clinically relevant deviations from normative performance or reporting residual symptoms were excluded. This stringent screening strategy ensured that the control group represented a recovered post-COVID population without ongoing symptomatology, rather than asymptomatic or mildly affected PCS cases.

We recruited participants through King's College Hospital NHS Trust and surrounding community networks. Eligible individuals were aged 16–65 years, fluent in English, and capable of providing informed consent. We excluded individuals with a history of major neurological or psychiatric disorders, current substance misuse, systemic or central nervous system inflammatory conditions, chronic respiratory or cardiac disease, recent vaccination (within two weeks), pregnancy or breastfeeding, BMI > 30 , or current use of immunomodulatory or psychoactive medications. We used the global severity score of Fatigue Assessment Inventory (FAI) to pre-screen participants for symptom severity (Schwartz et al., 1993). Individuals scoring greater than 4 on the global fatigue severity scale were classified into the PCS group with persistent fatigue, while those scoring 4 or below were included as recovered

controls. The threshold of >4 was selected to ensure inclusion of individuals with clinically meaningful levels of fatigue, consistent with prior applications of the FAI in post-viral and chronic fatigue contexts (Schwartz et al., 1993). Groups were balanced for age, sex, BMI, and acute COVID-19 severity. All participants underwent a structured diagnostic assessment using the Mini International Neuropsychiatric Interview (MINI) to exclude current or past major psychiatric disorders, including mood, anxiety, psychotic, and substance use disorders.

The study received ethical approval from the UK Health Research Authority (IRAS ID: 308661) and was conducted in accordance with the Declaration of Helsinki and Good Clinical Practice (GCP) guidelines. All participants provided written informed consent prior to study procedures.

We conducted an a priori power analysis using G*Power (version 3.1) to estimate the required sample size for detecting group differences using analysis of covariance (ANCOVA). Assuming a medium to large effect size (Cohen's $f = 0.30$, equivalent to $\eta^2 \approx 0.08$), $\alpha = 0.05$, power $(1 - \beta) = 0.80$, and three covariates (age, sex, and handedness), the estimated sample size required per group was 19. Our sample of 20 participants per group therefore provides adequate power to detect effects in this range. Importantly, individualized deviation scores were derived using the CentileBrain normative modelling framework, which is trained on large independent reference datasets comprising tens of thousands of healthy individuals across the lifespan. The normative model is estimated externally and independently of the present sample. Each participant's cortical thickness, surface area, or subcortical volume is compared against population-level expectations adjusted for age, sex, and intracranial volume, yielding subject-specific deviation z-scores. Thus, the estimation of individual deviations does not depend on the size of the current patient group. Sample size primarily affects the precision of group-level inference and the stability of spatial overlap maps, rather than the validity of deviation computation itself.

2.2. Procedures

Each participant completed a single study visit that included clinical assessments, venous blood collection, and a multimodal MRI scan. Clinical assessments included medical and psychiatric history, physical examination, vital signs, and a structured battery of standardized questionnaires. Fatigue was assessed using both the full Fatigue Assessment Inventory (Schwartz et al., 1993) and the Multidimensional Fatigue Inventory (Smets et al., 1995) to differentiate between mental and physical fatigue. Symptoms of autonomic dysfunction were evaluated using the Composite Autonomic Symptom Score (COMPASS-31) (Sletten et al., 2012), while mood and anxiety were measured using the Hospital Anxiety and Depression Scale (HADS) (Zigmond and Snaith, 1983). Sleep quality was assessed with the Pittsburgh Sleep Quality Index (Buysse et al., 1989), and respiratory symptoms were rated using the MRC Dyspnoea Scale (Bestall et al., 1999). Additional measures included the DePaul Symptom Questionnaire (Sunnquist et al., 2019), the American College of Rheumatology (ACR) fibromyalgia criteria (Wolfe et al., 2016), the PAIN Detect questionnaire (Shaygan et al., 2013), and Montreal Cognitive Assessment (Nasreddine et al., 2005). Most self-report questionnaires were completed remotely via the Qualtrics platform within 72 h of the in-person visit.

2.3. Blood analysis

Venous blood samples (up to 30 mL) were collected on the day of imaging. Standard hematological and biochemical panels - including erythrocyte sedimentation rate (ESR), C-reactive protein (CRP), thyroid-stimulating hormone (TSH), free thyroxine (T4), and liver function tests - were processed by Synnovis (Viapath) at King's College Hospital NHS Foundation Trust. Serum samples were frozen at -80°C and later assayed for cytokines and glial markers using electrochemiluminescence immunoassays (Meso Scale Discovery, Rockville, MD, USA). The

cytokine panel included interleukin-1 beta (IL-1 β), interleukin-2 (IL-2), interleukin-4 (IL-4), interleukin-6 (IL-6), interleukin-8 (IL-8), interleukin-10 (IL-10), interleukin-12p70 (IL-12p70), interleukin-13 (IL-13), tumor necrosis factor-alpha (TNF- α), and interferon-gamma (IFN- γ). Glial markers included glial fibrillary acidic protein (GFAP), measured with the ultrasensitive kit, and S100 β . Assays were performed in duplicate. Values below the detection limit were discarded. This led us to discard all quantifications of IL-2, IL-4 and IL-12p70. Outliers greater than three standard deviations from the group mean were winsorized. No analyte had more than 10% missing data, therefore all data were included.

2.4. Cognitron

Cognitive performance was assessed online using a battery of computerized tasks from the Cognitron platform, covering delayed verbal memory, working memory, executive function, attention, motor coordination, and spatial manipulation (Hampshire et al., 2024). Tasks included Verbal Analogies (abstract reasoning and semantic integration), Immediate and Delayed Prospective Memory (episodic memory and planning), 2D Spatial Manipulation (visuospatial working memory and mental rotation), Motor Control (sensorimotor coordination and reaction consistency), Spotter (vigilance and sustained attention), and Block Reasoning (fluid intelligence and rule-based problem solving). Each task involves trial-level response collection, capturing both accuracy and reaction time. Performance metrics were transformed into standardized deviation-from-expected (DFE) scores using normative models derived from a large independent sample ($n > 20,000$), adjusted for age, sex, handedness, and harmonized ethnicity. DFE scores represent standardized residuals (observed minus expected performance divided by normative standard deviation) and were computed separately for accuracy and reaction time.

2.5. Neuroimaging acquisition and processing

Magnetic resonance imaging (MRI) data were acquired using standardized imaging protocols on a 3T MRI scanner equipped with a standard head coil. High-resolution structural images were collected, including T1-weighted anatomical scans obtained using an MPRAGE sequence (isotropic voxel size = 1 mm^3 , TR = 2300 ms, TE = 2.98 ms, flip angle = 9° , field-of-view [FOV] = 256 mm) and complementary T2-weighted images (voxel size = 1 mm^3 , TR = 3200 ms, TE = 408 ms, flip angle = 120° , FOV = 256 mm). These sequences allowed detailed assessment of cortical thickness, cortical surface area, and subcortical brain volumes, leveraging the complementary tissue contrast provided by both imaging modalities. Image preprocessing and morphometric analyses were conducted using validated, automated pipelines implemented in FreeSurfer (version 7.2) (Fischl 2012). T1-weighted images were initially processed through skull stripping to remove non-brain tissue, followed by bias field correction to mitigate intensity inhomogeneities. Subsequently, images underwent segmentation into gray matter, white matter, and cerebrospinal fluid. The complementary T2-weighted images were integrated into the FreeSurfer pipeline, enhancing the accuracy of pial surface delineation and improving segmentation quality, particularly in cortical regions and areas adjacent to cerebrospinal fluid.

Rigorous quality control (QC) procedures were implemented at multiple stages of the neuroimaging pipeline to ensure the accuracy and reliability of morphometric estimates. In addition to the default FreeSurfer QC procedures, we utilized the open-source FSQC tool developed by the Deep-MI initiative (<https://deep-mi.org/fsqc/dev/index.html>) to automate the detection of potential segmentation errors and anatomical outliers across multiple FreeSurfer outputs (Bedford et al., 2023). Each participant's imaging data and segmentation outputs were subsequently subjected to detailed manual visual inspection by an experienced rater, using the FSQC interface alongside FreeSurfer's built-in visualizations.

Key features examined included motion artifacts, intensity non-uniformities, accuracy of white and pial surface placement, and the fidelity of cortical and subcortical parcellation. Datasets failing initial QC underwent manual correction, including adjustments to brain masks, removal of skull-strip artifacts, and refinements of white matter and pial boundaries. Corrected images were reprocessed through the FreeSurfer pipeline and re-evaluated to confirm improved segmentation quality. Only datasets that passed both automated and manual QC thresholds were retained for subsequent statistical analyses. This stringent multi-step approach ensured high-quality morphometric data and minimized the inclusion of artifacts or segmentation errors that could confound downstream individual-level deviation estimates or group-level comparisons.

2.6. Normative modelling approach

To quantify individual deviations in brain structure, we employed the *CentileBrain* normative modelling framework, which estimates expected neuroanatomical features - cortical thickness, surface area, and subcortical volumes - based on a large reference dataset of healthy controls using Gaussian Process Regression (Ge et al., 2024). Individual deviation scores were expressed as z-scores, reflecting how much a given regional brain measurement deviated from the normative prediction for that person's age, sex, and intracranial volume. We defined extreme deviations as z-scores falling outside the ± 1.96 threshold, corresponding to values in the top or bottom 2.5% of the normative distribution (i. e., below the 5th percentile or above the 95th percentile). For each cortical or subcortical region, we determined whether a participant exhibited an extreme deviation. This binarized information was used to calculate regional overlap, defined as the proportion of participants in each group (PCS or controls) showing extreme deviations in each region. Group differences in overlap were visualized as difference maps, where positive values indicated greater overlap in PCS and negative values indicated greater overlap in controls.

The *CentileBrain* normative model used in this study was previously trained on large independent reference datasets comprising tens of thousands of healthy individuals across the lifespan and was not recalibrated using the present sample. Model hyperparameters and Gaussian Process Regression settings were fixed prior to application. All participants in the current study (PCS and recovered controls) were processed identically through FreeSurfer and subsequently entered into the pre-trained normative framework. For each individual, region-specific expected morphometric values were generated conditional on age, sex, and intracranial volume, and deviation scores were calculated as standardized residuals (observed minus predicted divided by predictive uncertainty). The 20 recovered controls in this cohort were not used to train or fine-tune the normative model; instead, they served as an independent comparison group for evaluating deviation distributions between PCS and recovered individuals. No dataset-specific harmonization or recalibration procedures were performed. While deviation scores are defined relative to an external normative reference, group-level comparisons of deviation Z-scores were conducted to determine whether PCS participants exhibited systematic shifts in deviation burden or spatial distribution relative to recovered controls. These analyses operate on standardized residuals rather than raw morphometric measures and therefore assess differences in departure from population expectations, rather than classical case-control anatomical contrasts.

To examine whether these deviations converged within large-scale brain systems, we performed circuit-level overlap analyses. Following the principles of lesion-network mapping (Joutsa et al., 2022), for each individual, we also identified regions with extreme deviations and then mapped their structural connectivity profiles using a normative structural connectome derived from diffusion-weighted imaging data in the Human Connectome Project (Kruyer et al., 2024). For each deviated region, we extracted all directly connected regions (first-order neighbours) and combined them into a participant-specific circuit (Segal

et al., 2023). We then calculated the percentage of participants in each group who showed at least one extreme deviation within each circuit. Circuit-level overlap thus captured the frequency with which deviations occurred in anatomically connected subnetworks, even if the specific deviated nodes varied across individuals. Together, the regional and circuit-level analyses enabled us to assess whether the anatomical deviations observed in PCS were spatially random or showed consistent patterns of convergence within specific areas or networks. This multi-scale framework allowed us to identify not only which brain regions were most frequently affected across individuals, but also whether these deviations reflected disruption of shared neural circuits (Segal et al., 2023). Subcortical volumes were not included in the circuit-level overlap analysis because extreme deviations in subcortical regions were infrequent and did not differ meaningfully between groups, and because structural connectivity modelling within the adopted parcellation framework provides greater spatial reliability for cortical than subcortical nodes.

2.7. Network-diffusion modelling

To examine the potential propagation pathways of structural brain alterations across neural circuits, we implemented network-diffusion modelling using structural connectivity data from the Human Connectome Project (Chopra et al., 2023; Bhattarai et al., 2022; Agosta et al., 2025; Parsons et al., 2021; Shafiei et al., 2023) obtained from the ENIGMA toolbox. T1-weighted and diffusion-weighted images were preprocessed using standard HCP pipelines, including denoising, Gibbs ringing removal, eddy-current and motion correction, EPI distortion correction, and bias-field correction. Whole-brain probabilistic tractography was performed using MRtrix3 (tckgen) with constrained spherical deconvolution and anatomically constrained tractography (ACT). Streamlines were filtered using SIFT2 to improve quantitative correspondence with underlying white matter fiber densities. Nodes were defined according to the Desikan-Killiany atlas parcellation applied to T1 images, yielding 68 cortical regions and selected subcortical nuclei. Pairwise connection weights were computed as the SIFT2-weighted sum of streamlines connecting each node pair, normalized by node volume. The resulting symmetric connectivity matrix served as the structural constraint in the network diffusion models.

Subcortical volumes were not included in the network diffusion analyses because these models were specifically designed to examine spatial patterns of cortical thickness deviations. Cortical thickness is defined only for cortical regions and is parcellated according to the Desikan-Killiany cortical atlas used in the connectome framework. To ensure methodological consistency between the deviation maps and the structural connectivity matrix, diffusion modelling was therefore restricted to cortical nodes. This exclusion does not imply that subcortical structures are unimportant in post-COVID symptomatology; rather, in the present analytic framework, the diffusion process was defined over cortical thickness measures, which are inherently cortical metrics.

Our use of network-diffusion modelling is not intended to infer temporal or causal propagation of pathology. Instead, it tests whether the spatial distribution of cortical deviations is better explained by the topology of the structural connectome than by spatially constrained null models. The diffusion modelling was based on a heat-kernel diffusion process applied to a structural connectome derived from diffusion-weighted imaging as previously reported (Pandya et al., 2022), repeating the seeding process iteratively from each region of the parcellation used at a time. In this context, the term “seed” refers to regions whose simulated diffusion profiles best recapitulate the empirical spatial distribution of deviations under a connectome-constrained model. It should be interpreted as a statistical epicentre within the modelling framework, rather than as evidence of a temporally verified origin of pathology. The diffusion model involved optimizing the diffusion coefficient (β), which controls the rate of propagation within the connectome. Optimization of β was achieved by systematically evaluating

diffusion performance across a range of values (0.1–1.0, increments of 0.1), selecting the β parameter that maximized the correlation between the simulated diffusion pattern and empirical neuroanatomical thickness deviations as it was in this metric where we found most differences. Diffusion simulations were constrained to reflect propagation over a biologically plausible time range (0–24 months post-infection), enabling investigation of both short-term and long-term diffusion dynamics. Permutation testing was performed to assess the statistical significance of the observed diffusion patterns. Specifically, empirical diffusion results were compared against null distributions generated by running diffusion simulations on scrambled connectomes with randomized connectivity patterns and Euclidean distance-based surrogate connectomes. A total of 1000 permutations were conducted for each approach, establishing the statistical robustness of observed propagation patterns and identifying network hubs and pathways preferentially involved in structural alterations associated with prolonged fatigue following mild COVID-19 infection.

2.8. Molecular and cellular analyses

Regional microarray expression data: Gene expression data from the AHBA were obtained from six adult postmortem brains aged 24–57 years. Genetic probes were reannotated following standardized methods outlined in (Arnatkeviciute et al., 2019) to increase accuracy, and probes deemed unreliable were excluded. For each gene, the most stable probe was selected based on pooled donor correlations, resulting in a final dataset of 15,633 probes across the brain. Microarray samples were spatially assigned to the 83 regions of the Desikan-Killiany atlas40 using the abagen toolbox (<https://abagen.readthedocs.io/en/stable/>), with regional assignment based on corrected MNI coordinates (Markello et al., 2021). Assignments were constrained by hemisphere and cortical/subcortical boundaries to improve anatomical accuracy. Gene expression values were normalized using a robust sigmoid function and rescaled to a unit interval for cross-region and cross-donor comparability. Normalization was performed separately for cortical and subcortical regions to circumvent known differences in mean gene expression between cortex and subcortex. Due to limited availability of samples from the right hemisphere, we mirrored samples from the left hemisphere into the right hemisphere to ensure all regions are assigned values of gene expression for all genes.

2.9. Candidate gene analyses

To explore the molecular vulnerability of specific brain regions to SARS-CoV-2-related pathology, we examined whether regional variations in cortical thickness and surface area—derived from normative modelling - corresponded spatially with the constitutive expression of genes implicated in viral entry and neuroimmune signalling, as previously mapped using the same anatomical parcellation. We focused on a set of a priori candidate genes with well-established roles in mediating SARS-CoV-2 infection and its downstream inflammatory responses. TMPRSS2 (transmembrane serine protease 2) plays a critical role in priming the viral spike (S) protein, facilitating fusion of the viral and host cell membranes and enabling entry into host cells. TMPRSS2 is expressed in both respiratory and neural tissues and is thought to modulate region-specific susceptibility to direct viral invasion (Hoffmann et al., 2020). FURIN encodes a proprotein convertase that cleaves and activates the spike protein at the S1/S2 site, a step essential for viral infectivity. Its widespread expression in the central nervous system may contribute to facilitating viral spread in brain tissue, particularly in regions with high baseline FURIN activity (Coutard et al., 2020). NRP1 (neuropilin-1) has been identified as a co-receptor that enhances SARS-CoV-2 cell entry by stabilizing spike protein interactions with ACE2. Beyond its role in viral infectivity, NRP1 is also involved in axonal guidance and neurovascular signaling, raising the possibility that its expression could influence both entry pathways and downstream

neurovascular dysfunction (Cantuti-Castelvetri et al., 2020). TLR4 (toll-like receptor 4) is a key component of the innate immune system that recognizes pathogen-associated molecular patterns (PAMPs) and initiates inflammatory cascades. Recent studies have shown that TLR4 may directly respond to the SARS-CoV-2 spike protein, leading to exaggerated pro-inflammatory responses. In the CNS, TLR4 activation has been implicated in microglial priming and neuroinflammation, which are candidate mechanisms for post-viral neuropsychiatric sequelae (Shirato and Kizaki, 2021). Together, these genes represent plausible molecular entry points and immunological amplifiers of SARS-CoV-2-related effects on the brain. Unfortunately, the expression data for ACE2 did not pass our quality filtering criteria even if it constitutes the main point of entry for viral infection of cells and would therefore, at least theoretically, configure an important candidate gene in these analyses. Spatial correlations between their expression profiles and structural brain deviations may help to explain region-specific vulnerability and support the hypothesis that SARS-CoV-2-related neuropathology is shaped by underlying molecular architecture.

2.10. Cellular maps

To investigate the cellular substrates underlying the regional brain structural deviations observed in post-COVID syndrome, we derived cortical and subcortical cell-type distribution maps using the Britegea package (McKenzie et al., 2018). This analytical framework utilizes gene expression data from the Allen Human Brain Atlas (AHBA) and integrates curated gene sets derived from human single-cell transcriptomic studies to estimate the relative abundance of specific cell types across brain regions. The cell populations included in the analysis encompassed neurons, astrocytes, microglia, oligodendrocytes, oligodendrocyte precursor cells (OPCs), and endothelial cells. Gene markers corresponding to each cell class were used to compute regional enrichment scores, which were then aggregated and spatially aligned to the same Desikan-Killiany atlas used for our cortical thickness and surface area analyses. This ensured that cell-type density estimates were matched to the spatial resolution of our imaging data. We then assessed the relationship between cell-type densities and neuroanatomical deviations derived from the normative modelling framework. Specifically, we calculated region-wise correlations between cell-type density maps and z-scored deviations in cortical thickness and surface area. This approach would allow us to identify which cellular populations most strongly aligned with the pattern of structural brain changes observed in PCS, which in turn might hint at possible pathophysiological processes, including neuroinflammation (via microglial and astrocytic involvement), demyelination or oligodendrocyte dysfunction, and disruptions in neurovascular integrity.

2.11. Neurotransmitter molecular targets

To further investigate the molecular architecture associated with regional brain alterations, we incorporated receptor density maps derived from normative Positron Emission Tomography (PET) imaging data. These templates represent in vivo estimates of neurotransmitter receptor and transporter availability across the human brain and were sourced from publicly available datasets generated using high-affinity radioligands in healthy adult populations and are available as part of the *neuromaps* toolbox (Markello et al., 2022). The receptor maps included distributions for key molecular systems implicated in neuropsychiatric function and fatigue regulation, such as the serotonergic system (5-HT1A, 5-HT1B, 5-HT2A, 5-HT4, 5-HT6, 5HT transporter), dopaminergic system (D1, D2, and dopamine transporter [DAT]), glutamatergic system (NMDA, mGluR5), cholinergic system (M1 muscarinic receptor, vesicular acetylcholine transporter [VACHT]), cannabinoid system (CB1 receptor), GABAergic system (GABA-A receptor), opioid system (μ , κ , and δ opioid receptors), noradrenergic system (norepinephrine transporter [NET]), and histaminergic

system (H3 receptor). For details on the data originating each of these maps, we direct the reader to the original *neuromaps* publication (Markello et al., 2022). Each PET-derived receptor density map was spatially registered and parcellated to match the Desikan-Killiany atlas used for cortical morphometry analyses. This alignment enabled region-wise comparison between individual structural deviations (e.g., cortical thickness or surface area z-scores from normative modelling) and normative receptor distributions. Associations were quantified using correlation and canonical correlation analyses, allowing us to assess which neurochemical systems best predicted the spatial distribution of observed structural alterations in PCS. This receptor-informed analysis would provide a neurochemical dimension to the observed anatomical deviations, facilitating the prioritisation of specific molecular systems that may underlie or modulate the persistent fatigue and neuropsychiatric symptoms reported in post-COVID syndrome.

2.12. Statistical analysis

Statistical analyses were performed to examine group differences and associations across demographic, clinical, cognitive, and neuroimaging measures. Between-group comparisons for continuous demographic and clinical variables were conducted using independent-samples t-tests, with effect sizes reported as Cohen's d. Categorical variables were compared using chi-square tests. Bayes factors (BF_{10}) were also computed to quantify the strength of evidence supporting group differences. For neuroimaging data, all group comparisons - including regional structural deviations (z-scores for cortical thickness, surface area, and subcortical volumes) - were evaluated using analysis of covariance (ANCOVA) models. These models included age, gender, handedness (manual dexterity), and total intracranial volume (TIV) as covariates to control for known sources of anatomical and physiological variability. To evaluate relationships between regional neuroanatomical deviations and molecular or cellular variables (candidate gene expression, cellular densities, receptor distributions), Spearman's rank correlation analyses were conducted. Given the inherent spatial autocorrelation in cortical neuroimaging data, we employed permutation-based spin tests (1000 permutations), generating null distributions by systematically rotating cortical maps while preserving spatial relationships (Vasa method) (Vasa et al., 2018). This rigorous approach ensured accurate assessments of statistical significance and minimized false-positive findings due to spatial dependencies. For multivariate integration of structural imaging metrics with molecular and cellular profiles, canonical correlation analysis (CCA) was performed (Mihalik et al., 2022). Statistical significance of canonical correlations was assessed through permutation testing (1000 permutations), also accounting for spatial autocorrelation using spin permutations. Network-diffusion modelling outcomes were validated against null models generated by permutations of structural connectivity matrices (scrambled connectomes) and Euclidean-distance-based surrogate connectomes (1000 permutations), further ensuring the robustness and specificity of observed diffusion patterns (Vasa and Misić, 2022). All statistical procedures and significance testing were performed using Python and R software packages specialized for neuroimaging and spatial analyses, with an alpha threshold set at $p < 0.05$ (two-tailed) for statistical significance, unless otherwise specified.

3. Results

3.1. Sociodemographics and clinical characterisation

Results have been first reported elsewhere. Here we describe the main findings that might be relevant to interpret the new neuroimaging findings. The groups were matched for age, sex, ethnicity, and body mass index (BMI), with no significant differences observed across these variables (all $p > 0.5$; Table 1). However, groups differed significantly in years of education, with HC participants reporting higher educational

Table 1

Sociodemographic and clinical characteristics of PCS and healthy control participants. Group comparisons of demographic and COVID-19-related variables between individuals with Post-COVID-19 Syndrome (PCS) and matched healthy controls (HC). Data are shown as mean (standard deviation) or counts. Between-group differences were assessed using independent samples t-tests or chi-squared tests as appropriate. Bayes factors (BF_{10}) quantify evidence for group differences; values > 1 indicate increasing support for the alternative hypothesis. Asterisks (*) denote statistically significant results at $p < 0.05$ or $BF_{10} > 1$.

Variable	PCS Mean (std)	HC Mean (std)	Statistics	p	B_{10}
Age	40.6 (10.9)	40.0 (10.4)	T(38) = 0.208	0.837	0.312
Gender	18F/2M	17F/3M	χ^2 (Nalbandian et al., 2021) (1) = 0.229	0.663	N/A
Ethnicity	White: 18 Mixed: 0 Asian: 1 Black: 0 South Asian: 1 Hispanic: 0	White: 15 Mixed: 1 Asian: 1 Black: 2 South Asian: 0 Hispanic: 1	χ^2 (Nalbandian et al., 2021) (5) = 5.273	0.384	N/A
BMI	23.8 (3.44)	24.4 (3.84)	T(38) = -0.55	0.582	0.346
Years of education	17.30 (2.94)	19.35 (3.13)	T(38) = -2.13	0.039*	1.845
Time since first infection (months)	23.0 (9.17)	15.7 (11.5)	T(38) = 2.21	0.033*	0.749
Vaccination prior infection	Yes: 4 No: 16	Yes: 14 No: 6	χ^2 (Nalbandian et al., 2021) (1) = 10.101	0.001*	N/A

attainment. PCS participants also had a significantly longer duration since first SARS-CoV-2 infection, and were significantly less likely to have been vaccinated prior to infection. Clinically, individuals with PCS reported significantly greater symptom burden across multiple domains. These included fatigue (both visual analogue scale and multidimensional components), sleep disturbance (PSQI), post-exertional malaise, autonomic dysfunction (COMPASS-31), affective symptoms (HADS depression/anxiety), PTSD-related symptoms, and musculoskeletal complaints (WPI and SS scores). Notably, 40% of PCS participants met criteria for fibromyalgia, compared to none in the HC group. These differences were supported by both frequentist ($p < 0.05$) and Bayesian inference, with many outcomes yielding strong to extreme evidence in favor of group separation ($BF_{10} > 10$), particularly for fatigue, post-exertional malaise, and pain sensitivity (Supplementary Table S1).

3.2. Clinical routine blood tests

Group comparisons of routine blood parameters revealed no major differences between PCS and HC participants (Supplementary Table S2). However, PCS participants showed numerically higher creatinine and Gamma-GT levels, with corresponding Bayes factors ($BF_{10} = 1.232$ and 1.735 , respectively) providing anecdotal to moderate evidence for group differences. Platelet count also trended higher in the PCS group ($p = 0.08$; $BF_{10} = 1.123$). Other metabolic, inflammatory, and haematological markers - including C-reactive protein, white and red cell counts, haemoglobin, and thyroid function - did not differ significantly between groups.

3.3. Serum cytokines and glial markers

No statistically significant differences were observed in circulating levels of IFN- γ , IL-1 β , IL-6, IL-8, IL-10, IL-13, TNF- α or GFAP (Supplementary Table S3). However, TNF- α ($p = 0.116$; $BF_{10} = 1.094$)

and S100 β ($p = 0.149$; $BF_{10} = 1.025$) showed suggestive evidence of elevation in the PCS group, which may be compatible with subtle glial or immune-related processes, although these findings were not statistically robust.

3.4. Cognitive performance

Cognitive testing revealed modest and domain-specific group differences between groups (Supplementary Table S4). PCS participants showed significantly poorer performance on the delayed object memory task (RT_DFE; $p = 0.02$; $BF_{10} = 2.653$), as well as trend-level impairments in the Lead Balloon task (abstract reasoning; $p = 0.06$; $BF_{10} = 1.330$) and vigilance (Spotter task RT; $p = 0.06$; $BF_{10} = 1.303$). Other domains - including motor control, 2D spatial manipulation, and verbal analogies - did not show significant differences. Overall, Bayesian analysis provided moderate evidence for reduced memory and attentional function in PCS.

3.5. Group differences in the total amounts of positive and negative extreme deviations

Using our predefined threshold of $|z| > 1.96$, extreme deviations were mostly present for thickness and absent for surface and subcortical volume. PCS and control participants did not differ in the total number of either positive or negative extreme deviations in cortical thickness (positive: $F(35) = 0.082$, $p = 0.776$; negative: $F(35) = 0.368$, $p = 0.548$) (see Fig. 1).

3.6. Links between total amounts of positive and negative extreme thickness deviations and clinical, blood and cognitive parameters

Exploratory partial spearman correlations accounting for group, age, gender, dexterity and TIV revealed significant positive correlations between the total amount of positive extreme thickness deviations and performance in the Spotter task and blood magnesium, and negative correlations with AST levels, mental fatigue and total FAI scores, as well as severity of autonomic and immune symptoms assessed by the DePaul questionnaire. In the case of the total amount of negative extreme thickness deviations we found positive associations with serum S100 β ,

blood creatinine and urea, and blood count of eosinophils. None of these associations would have survived FDR correction for the total number of parameters assessed (Supplementary Table S5) and should therefore be interpreted cautiously.

3.7. Group differences in regional structural deviation Z-scores

Normative modelling revealed subtle but spatially distinct patterns of cortical deviation in PCS. At the group level, patients exhibited reduced cortical thickness deviation Z-scores in the left medial and right lateral orbitofrontal cortex, right pars opercularis and triangularis of the frontal cortex and left fusiform gyrus; and increased deviation Z-scores thickness in the left laterocipital cortex and right post-central gyrus (Fig. 2). Surface area Z-score deviation scores showed reduced values in the right postcentral gyrus and increases in the right frontal pole. None of these regions would have survived FDR correction for the number of regions tested. For subcortical volumes, no comparisons reached significance.

As a sensitivity analysis and because alterations in brain structure in PCS are often simply attributed to the presence of olfactory dysfunction, we repeated the same group comparisons on deviation Z-scores this time additionally accounting for the presence of anosmia/ageusia. As shown in Supplementary Fig. S1, many of the reported group differences were still evident and others not reaching significance before did after this additional covariate. For cortical thickness, we found decreases in Z scores for left fusiform and middle orbitofrontal gyri and an increase for the right isthmus cingulate. For surface area, we found increases in the left caudal middle frontal gyrus, left pars orbitalis and right pars triangularis of the frontal gyrus, and right frontal pole, and decreases in the right postcentral gyrus (Supplementary Table S6).

3.8. Regional and circuit overlap of extreme positive and negative deviations

Extreme deviations were heterogeneous at the individual regional level with every individual region being identified as extreme in <35% of the members of either group. However, a circuit-based overlap analysis, where we considered deviated regions as well as all other brain regions structurally connected to them, revealed significant convergence within specific structural networks, affecting up to an excess of 50% of patients in the PCS group. These deviations mapped predominantly onto salience and attention-related circuits, suggesting distributed but coherent alterations in anatomically and functionally connected regions (Fig. 3).

3.9. Network-diffusion modelling

Network diffusion modelling demonstrated that structural deviations in PCS are not randomly distributed but follow pathways consistent with structural connectivity. The optimized model ($\beta = 0.010$) produced simulated deviation patterns that significantly correlated with empirical data (Fig. 4). Posterior-temporoparietal regions emerged as likely “seeds” for network-wide propagation, showing the highest seed likelihood scores. These regions are involved in multisensory integration and attention and may represent key vulnerability nodes in PCS. Permutation testing confirmed that these findings exceeded chance expectations ($p < 0.05$), with simulations based on scrambled connectomes and Euclidean distance metrics showing significantly lower correspondence with empirical data.

3.10. Candidate genes related to SARS-CoV-2 entry into host cells

Spatial correlations between neuroanatomical deviations and candidate gene expression revealed strong associations with TMPRSS2, a known facilitator of SARS-CoV-2 entry into host cells. TMPRSS2 expression was significantly associated with regional changes in cortical

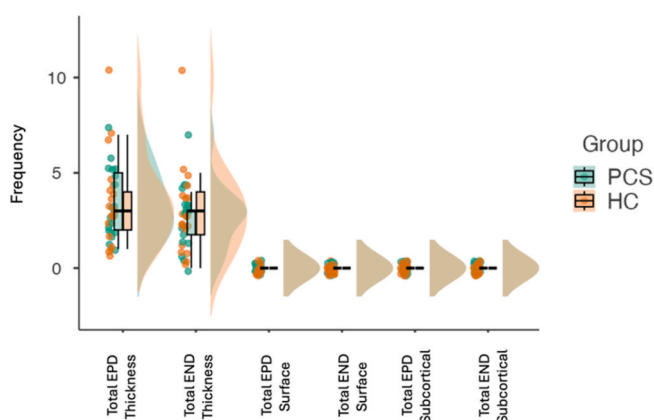


Fig. 1. Distribution of Individual-Level Extreme Neuroanatomical Deviations in PCS and Healthy Controls. Violin plots showing the distribution of total extreme deviations in cortical and subcortical morphometry metrics by group (PCS vs HC). Each plot displays the number of extreme deviations per participant (z -scores $|z| > 1.96$) across six categories: positive and negative deviations in cortical thickness, surface area, and subcortical volume. Violin widths reflect kernel density estimates, with overlaid boxplots showing medians and interquartile ranges. Individual data points are also plotted. PCS = post-COVID syndrome group; HC = healthy controls; EPD – extreme positive deviations; END – extreme negative deviations.

Post-COVID19 vs Controls (accounting for age, gender, dexterity and TIV)

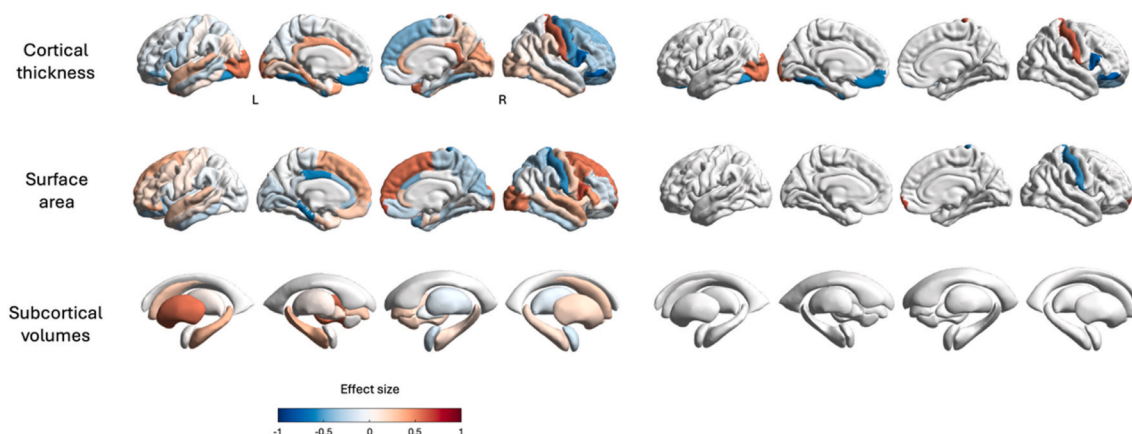
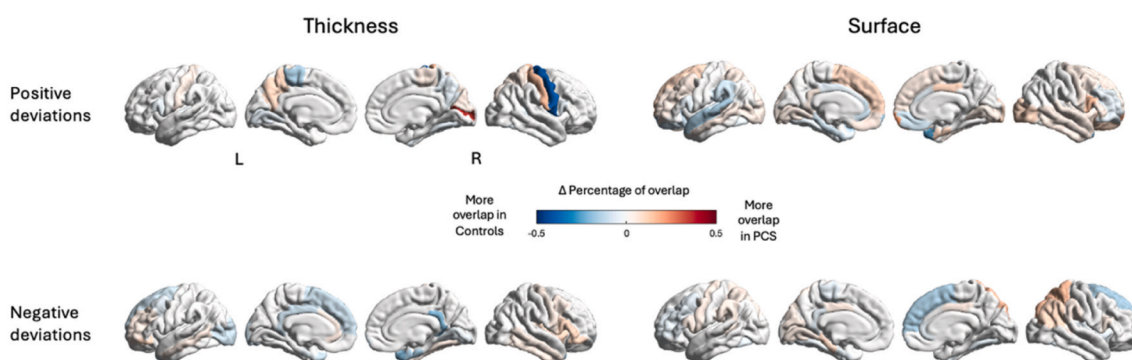


Fig. 2. Regional differences in cortical thickness, surface area, and subcortical volumes deviation scores between post-COVID-19 patients and healthy controls. Comparisons were adjusted for age, gender, handedness, and total intracranial volume (TIV). Colours indicate effect sizes (Cohen's d), with red representing increased measures and blue indicating decreased measures in post-COVID-19 patients compared to controls. Significant results are thresholded at uncorrected $p < 0.05$. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

A. Considering only regional deviation scores



B. Considering regional deviation scores and all regions structurally connected with deviated regions

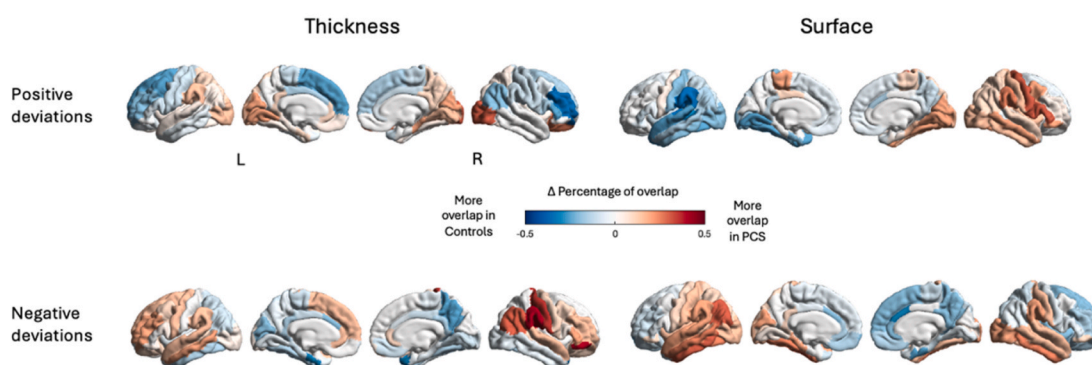


Fig. 3. Regional (A) and circuit-based (B) overlap analysis comparing positive and negative deviations in cortical thickness and surface area between post-COVID syndrome (PCS) patients and healthy controls. Colour gradients indicate the difference (Δ percentage overlap) in the proportion of participants showing deviations, with red reflecting greater overlap in PCS patients and blue representing greater overlap in controls. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

thickness, surface area, and seed likelihood from network diffusion models (Fig. 5). Other entry-related genes, such as *FURIN*, *NRP1*, and *TLR4*, showed weaker or nonsignificant associations.

3.11. Cellular and neurochemical correlates

Canonical correlation analyses (CCA) identified significant relationships between brain structural deviations and regional cell-type distributions. The strongest loadings involved neurons and microglia,

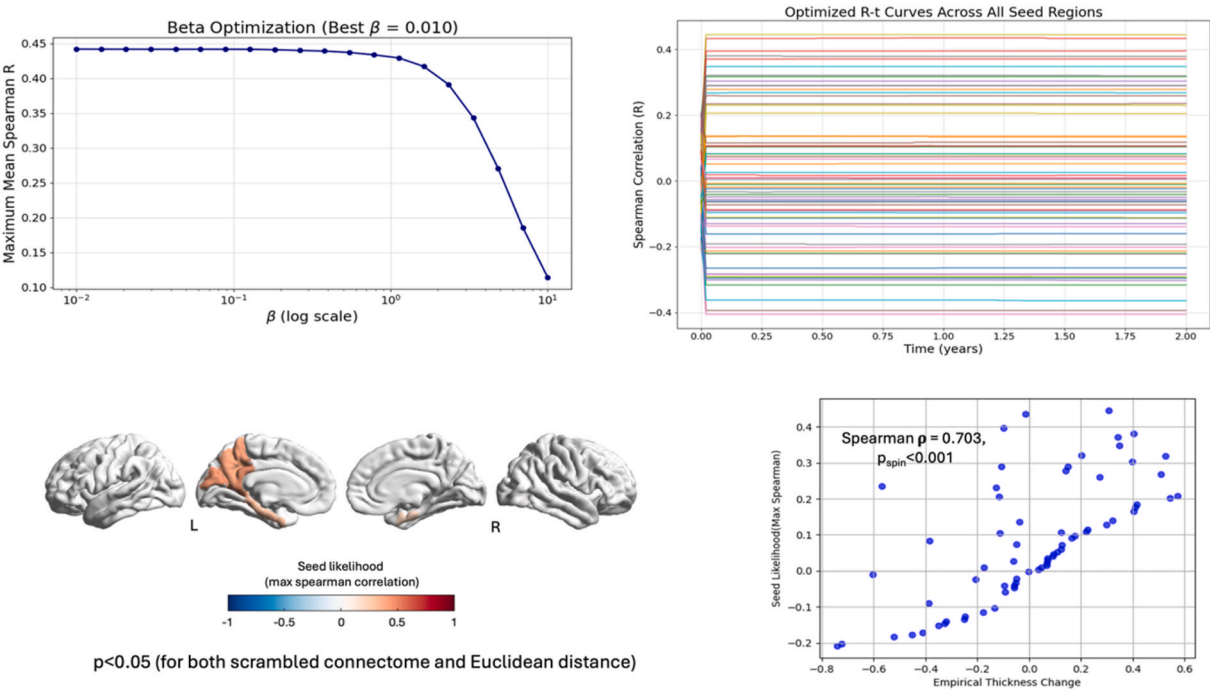


Fig. 4. Network diffusion modelling results. Upper left panel shows beta optimization, identifying the best diffusion coefficient ($\beta = 0.010$) based on maximum mean Spearman correlation. Upper right panel illustrates optimized Spearman correlation (R) curves over time across all potential seed regions. The lower left panel maps seed likelihood (maximum Spearman correlation) across brain regions, with higher likelihood regions indicated in red. The lower right panel demonstrates the significant positive correlation between seed likelihood and empirical cortical thickness changes (Spearman $\rho = 0.703$, $p < 0.001$). Permutation testing against scrambled connectomes and Euclidean distance models confirmed the significance of these findings ($p < 0.05$). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

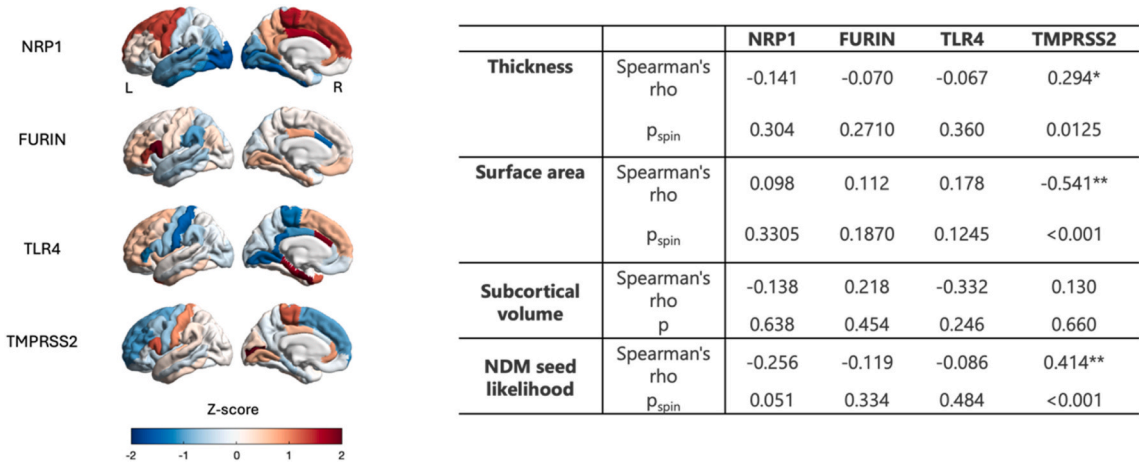


Fig. 5. Associations between SARS-CoV-2-related gene expression (NRP1, FURIN, TLR4, TMPRSS2) and neuroimaging metrics (cortical thickness, surface area, subcortical volume, and network diffusion model [NDM] seed likelihood). Left panel illustrates regional mRNA expression (Z-scores). Right panel summarizes Spearman's correlations (ρ) and associated permutation-based spin-test p-values (p_{spin}), highlighting significant associations (* $p < 0.05$, ** $p < 0.001$).

suggesting a role for neuroinflammatory and excitatory/inhibitory imbalances in PCS-related alterations (Fig. 6A). Associations were stronger for cortical thickness than surface area, consistent with prior work showing thickness as a sensitive index of neuroimmune changes. Receptor-based CCA further linked cortical deviations in thickness with a multivariate pattern of neurotransmitter systems. The most robust negative associations were observed for serotonergic 5HT1A and B, 5HT6, cholinergic (VACHT), and cannabinoid (CB1) targets, while mGlu5R had the highest positive loading (Fig. 6B). The neurotransmitter system CCA model did not reach significance for deviation in surface area.

4. Discussion

We applied a multimodal analytic framework to investigate neuro-anatomical deviations in individuals with PCS and persistent fatigue following mild SARS-CoV-2 infection. By integrating individualized morphometric deviation mapping with gene expression profiles, cellular composition, receptor distributions, and network diffusion modelling, we identified spatially organized patterns of cortical change. Although individual regional deviations were modest and variable, they converged across structurally connected circuits and aligned with specific molecular and neurochemical features. While preliminary, these

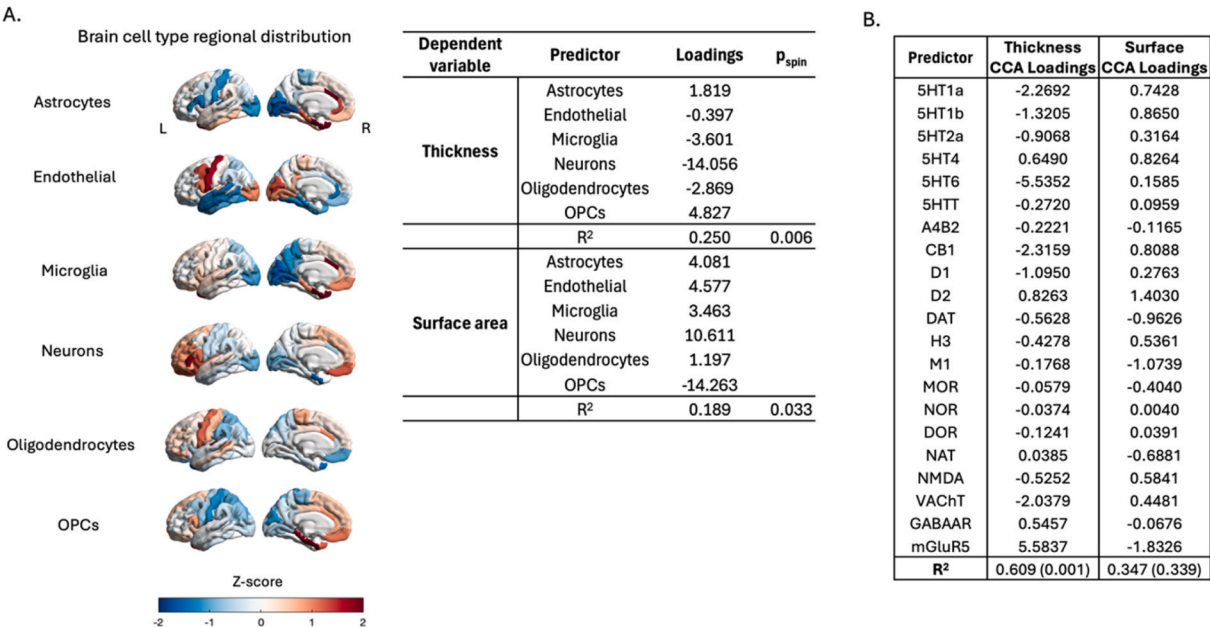


Fig. 6. Cellular and neuroreceptor correlates of structural deviations in PCS. (A) Brain cell-type regional distributions illustrating Z-scores for astrocytes, endothelial cells, microglia, neurons, oligodendrocyte, and oligodendrocyte precursor cells (OPCs). The accompanying table summarizes canonical correlation analysis (CCA) loadings and permutation-based spin-test p-values (p_{spin}), indicating significant associations between specific cell types and regional alterations in cortical thickness and surface area. (B) Canonical correlation analysis loadings for neuroreceptor densities predicting cortical thickness and surface area alterations. Predictors include serotonergic (5HT receptors), cannabinoid (CB1), dopaminergic (D1, D2, DAT), histaminergic (H3), cholinergic (M1, VACHT), glutamatergic (NMDA, mGluR5), opioid (MOR, NOR, DOR), noradrenergic (NAT), amyloid-beta (A4B2), and GABAergic (GABAAR) systems. The table also reports R^2 values with corresponding permutation-based significance (p-values in parentheses), highlighting stronger associations for thickness compared to surface area.

findings lay the groundwork for future studies seeking to clarify the biological basis of post-COVID fatigue and identify potential targets for intervention.

We observed reduced cortical thickness in orbitofrontal regions and increased thickness in occipital and sensory cortices. Recent systematic syntheses of neuroimaging findings in acute and post-acute COVID-19 cohorts have reported heterogeneous but recurrent patterns of cortical thinning, white matter microstructural alterations, and network-level disruptions, particularly involving frontal, temporal, and limbic circuits (Li et al., 2025). These findings underscore both the variability and the distributed nature of COVID-related brain changes, while also highlighting the limitations of conventional group-level comparisons in capturing individual heterogeneity. Moreover, these changes align with prior evidence of frontal vulnerability in chronic fatigue and post-infectious conditions and may reflect disruption of circuits involved in reward processing, emotional regulation, and sensory integration (Douaud et al., 2022; Lu et al., 2020; Alhazmi et al., 2023). Importantly, deviation estimates were derived from a pretrained large-scale normative reference and were independent of the local control group, ensuring that individualized deviations reflect departure from population expectations rather than cohort-specific calibration. Rather than affecting the same regions uniformly, deviations clustered within structurally connected networks, suggesting that circuit-level organization better accounts for symptom-related alterations than isolated anatomical loci. Similar conclusions have been reached regarding brain structure and function within the context of other classical neuropsychiatric disorders (Segal et al., 2023). Because no individual regional morphometric findings survived correction for multiple comparisons, all downstream spatial correspondence analyses should be interpreted as exploratory analyses anchored on subtle spatial effect distributions rather than robust focal anatomical abnormalities.

Network diffusion modelling extended these findings by identifying posterior-temporoparietal regions as likely sources for a slow propagation of structural alterations across the connectome, reinforcing their role as integrative hubs for sensory and attentional information -

functions frequently disrupted in PCS (Davis et al., 2021; Chiappelli and Fotovat, 2022; Churchill et al., 2023, 2024). We optimized a diffusion coefficient ($\beta = 0.010$) within a biologically realistic range, indicating a slow and spatially constrained propagation pattern (Chopra et al., 2023; Bhattarai et al., 2022; Agosta et al., 2025; Parsons et al., 2021; Shafiei et al., 2023). The same regions also showed higher expression of TMPRSS2, a host gene required for SARS-CoV-2 cellular entry (Hoffmann et al., 2020). While TMPRSS2 expression alone does not imply ongoing viral effects, its spatial correlation with structural deviation patterns suggests that pre-existing molecular architecture may influence regional vulnerability during or after SARS-CoV-2 infection (Arnatkeviciute et al., 2022; Mroczek et al., 2021). It is worth noting though our use of diffusion modelling here is not intended to imply a temporally verified spreading process as in progressive neurodegenerative disease (Xia et al., 2025), but rather to assess whether the empirical spatial topology of deviations is statistically consistent with a connectivity-informed propagation model compared with spatially constrained nulls. In this sense, we leverage the diffusion framework as a topographical organizing principle - testing whether structural covariance of deviations respects the geometry of the connectome - rather than as a literal model of biological propagation over time.

We found that cortical thickness deviations correlated with normative maps of neuronal and microglial density. Regions with greater neuronal density may be more susceptible to deviation through increased metabolic or synaptic demands (Clemente-Suarez et al., 2023; Pulido and Ryan, 2021). Microglial alignment may reflect structural consequences of immune surveillance or low-grade inflammation (Chagas and Serfaty, 2024; Gerhard, 2023). These associations highlight potential cellular substrates of PCS-related cortical changes. In parallel, canonical correlation analyses linked structural deviations to the distribution of serotonergic, cholinergic, cannabinoid, and glutamatergic receptors. Regions enriched in 5-HT1A, 5-HT1B, 5-HT6, VACHT, CB1, and mGlu5 showed the strongest anatomical correspondence. These systems contribute to arousal, mood, cognition, and fatigue regulation (Leitzke et al., 2025; Wong et al., 2023; Eslami and Joshaghani, 2024;

Cinar et al., 2022; Meeusen and Roelands, 2018). While we did not assess receptor binding or neurotransmitter function directly, the spatial alignment with structural deviations indicates that cortical regions embedded in specific neuromodulatory environments might be differentially affected in PCS. These findings are particularly interesting within the context of ongoing pharmacological trials investigating neuromodulatory agents - such as vortioxetine, which modulates 5-HT1A, 5-HT1B, and 5-HT6 receptors (Stahl, 2015) - for the treatment of cognitive and affective symptoms in PCS (McIntyre et al., 2024).

PCS participants showed impairments in delayed visual memory and sustained attention, with slower reaction times on delayed object recall and trend-level reductions in vigilance performance. These cognitive differences showed partial spatial correspondence with the distributed structural deviation patterns identified in PCS. Notably, sustained attention performance correlated with the burden of extreme positive deviations in cortical thickness, suggesting that individuals showing greater anatomical deviation also exhibited measurable functional change (Hampshire et al., 2024; Wood et al., 2025). Although serum biomarkers showed no major group-level differences, exploratory analyses revealed associations between cortical thinning and S100 β , creatinine, urea, and eosinophils, consistent with subtle glial and systemic contributions to brain structure. These patterns converge on a model in which regional structural alterations, likely embedded within neuro-immune and neuromodulatory gradients, underlie selective cognitive dysfunction in PCS. Peripheral cytokines may nevertheless not reliably reflect central nervous system inflammatory activity, and the absence of strong associations between circulating markers, structural deviations, and cognition underscores the need for direct in vivo measures of neuroinflammation to clarify brain-immune coupling in post-COVID-19 syndrome. Moreover, emerging work has proposed biologically informed subtypes of COVID-19 and psychiatric conditions based on multimodal data integration (Wang et al., 2026). While our sample size does not permit formal subtype derivation, the combination of individualized normative deviation mapping with transcriptomic and receptor-informed spatial analyses represents a conceptual step toward mechanistically grounded stratification. Larger cohorts incorporating longitudinal immune profiling and multimodal imaging will be required to determine whether distinct PCS phenotypes map onto dissociable neurobiological signatures (Zhang et al., 2025; Fu et al., 2019).

Post-COVID syndrome is clinically heterogeneous, with variable combinations of fatigue, cognitive impairment, autonomic dysfunction, sleep disturbance, and affective symptoms. Although we observed subtle associations between structural deviation burden and selected cognitive measures, these exploratory correlations did not survive correction for multiple comparisons and should be interpreted cautiously. We did not identify robust region-specific mappings between distinct symptom dimensions and specific cortical deviations in this cohort. Rather than supporting a one-to-one regional-symptom correspondence, our findings suggest that structural deviations may reflect disruption of distributed circuits implicated in attention, interoception, and salience processing, which could manifest differently across individuals depending on baseline vulnerability, immune response, and compensatory capacity. Larger cohorts incorporating dimensional symptom modelling, latent class approaches, or longitudinal designs will be necessary to determine whether distinct PCS phenotypes map onto dissociable neurobiological signatures.

Our study's integrative approach - combining personalized structural modelling with gene expression and receptor data - represents a methodological advance and supports a biologically coherent narrative of PCS-related fatigue. More broadly, recent work integrating transcriptomic and neuroimaging data has emphasized the potential of multimodal frameworks to move beyond descriptive neuroimaging toward biologically informed stratification and treatment selection (Wang et al., 2024). While the present study was not designed for diagnostic prediction, the alignment between individualized structural deviation patterns and normative molecular gradients illustrates how such

approaches may, in the future, contribute to mechanistically informed phenotyping. Translation to clinical decision-making will require replication in larger cohorts, longitudinal validation, and integration with prospective treatment outcomes. Nevertheless, several limitations must be acknowledged. First, the sample size was small, limiting the detection of small effects, reducing the precision of multivariate associations, and constraining generalizability. Accordingly, our molecular, cellular, and network-level findings should similarly be considered exploratory and hypothesis-generating. Replication in larger, ideally longitudinal cohorts will be necessary to confirm the stability and clinical relevance of the reported patterns. Future larger-scale studies should evaluate the robustness of overlap patterns across alternative deviation thresholds and continuous burden metrics. Second, the cross-sectional design precludes inference about temporal progression or reversibility. For instance, longitudinal neuroimaging would be required to test whether posterior-temporoparietal regions indeed show earlier or progressive changes relative to other areas. Such data would allow formal testing of temporal precedence and dynamic spread. Third, while we adjusted for key demographic and imaging covariates, unmeasured confounders - such as variations in SARS-CoV2 variant and off-label treatment with supplements/hormonal therapy/histamine blockers - may have influenced both brain structure and symptom expression. Because education and vaccination status differed systematically between groups and were collinear with group membership, we did not include them as nuisance covariates in primary models risking model instability; larger cohorts with balanced sampling will be required to disentangle PCS-related effects from these correlated exposures. Moreover, unmeasured visual symptoms could contribute to variance in occipital and sensory regions but formal ophthalmic assessment was not available in this dataset. Fourth, molecular and receptor associations were based on population-average postmortem datasets - including the Allen Human Brain Atlas, which includes only six adult donors - limiting generalizability and precluding inferences about individual-level or state-dependent biology (Shen et al., 2012). The receptor maps derived from PET templates also reflect healthy population averages and do not capture potential dynamic changes in neurotransmitter function following infection/inflammation (Felger and Treadway, 2017). These datasets do not capture individual-level biological variability, state-dependent molecular changes, or dynamic alterations induced by infection or inflammation. Accordingly, the spatial correlations reported here should be interpreted as reflecting alignment with normative molecular architecture rather than direct evidence of altered gene expression or receptor density in PCS participants. Furthermore, while spatial permutation (spin) testing was used to account for spatial autocorrelation, these analyses remain correlational and cannot establish causal relationships between molecular gradients and structural alterations. Given the biological heterogeneity of SARS-CoV-2, including differences across viral variants and host immune responses, and the absence of variant-level stratification in the present cohort, the present findings should not be interpreted as demonstrating neurobiological specificity to PCS, and may instead reflect partially shared mechanisms across post-viral, fatigue-related, and neuroimmune conditions.

Taken together, our findings reveal that PCS-related fatigue is associated with distributed cortical deviations that follow structural connectivity and co-localize with specific molecular and cellular features. These results characterize PCS as a condition involving spatially structured brain alterations embedded within established neurochemical and immune architectures. Future research should aim to replicate and extend these findings in larger, longitudinal cohorts, ideally integrating multimodal imaging with direct measures of neuroinflammation and neurotransmitter function.

CRedit authorship contribution statement

Daniel Martins: Conceptualization, Data curation, Formal analysis,

Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Ziyuan Cai**: Formal analysis, Writing – review & editing. **Nicole Mariani**: Formal analysis, Writing – review & editing. **Alessandra Borsini**: Formal analysis, Funding acquisition, Resources, Writing – review & editing. **Valeria Mondelli**: Funding acquisition, Resources, Supervision, Writing – review & editing. **Brandi Eiff**: Data curation, Investigation, Writing – review & editing. **Silvia Rota**: Investigation, Project administration, Writing – review & editing. **Daniel van Wamelen**: Investigation, Writing – review & editing. **Timothy Nicholson**: Investigation, Writing – review & editing. **Laila Rida**: Software, Writing – review & editing. **Adam Hampshire**: Software, Writing – review & editing. **Lia Fernandes**: Validation, Writing – review & editing. **Fernando Zelaya**: Methodology, Writing – review & editing. **Aleksandra Podlowska**: Project administration, Writing – review & editing. **Ray Chaudhuri**: Project administration, Writing – review & editing. **Federico Turkheimer**: Conceptualization, Funding acquisition, Supervision, Validation, Writing – review & editing. **Steven C.R. Williams**: Conceptualization, Funding acquisition, Validation, Writing – review & editing. **Mattia Veronese**: Conceptualization, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was funded by the National Institute for Health and Care Research (NIHR) Maudsley Biomedical Research Centre (BRC), South London and Maudsley NHS Foundation Trust, under the “Reach Out” funding call (Ref: R0-01). We are grateful to all participants for their time and commitment to the study. We thank the radiographers and technical staff at the Centre for Neuroimaging Sciences (King's College London) for their assistance with MRI acquisition. We also acknowledge Synnovis (Viapath) for processing routine blood analyses, the Clinical Research Facility (CRF) team at King's College Hospital Foundation Trust for their support during participant visits and blood collection, and the Cognitron platform team for assistance with cognitive data management and normative modeling.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.bbih.2026.101292>.

Data availability

Data will be made available on request.

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